

Review of Previous Meeting

- **To-Do Items**

- **Single-frame & Sequence VLM Test:** Conduct detection tests using GPT/Gemini on both cherry-picked single frames and 41-frame temporal sequences. The sequence-based approach provides 1.34 seconds of movement data to verify if the models can identify glitches that require temporal context, such as speed-based anomalies or state-change defects that are unidentifiable in static images.
- **Expand Annotation Data:** Manually annotate at least 10 more WR videos to check if the glitch types and their durations are consistent across different runs.

Single-frame VLM Test

- **Results:** Correct labels obtained in only 2 out of 8 cases.
- **Observations:** Detection appears feasible when spatial or locational cues are prominent and distinct.
- **Limitation:** Static images fail to capture the temporal context essential for identifying speedrun-specific glitches.

Frame Sequence VLM Test

- **Results:** While only 3 out of 8 were labeled correctly, the reasoning for 6 out of 8 cases accurately captured the actual events.
- **Analysis of Failures:** The two cases that failed to be identified involved glitches characterized by extremely high-speed movement.
- **Hypothesis Testing:** I explicitly stated the 1.34-second duration in the prompt to improve detection ; however, this backfired as the model became biased by the numerical data, leading to a decrease in accuracy.

Expand Annotation Data

- **WR Progression:** The chart illustrates the historical trend of World Records (WR) for the SM64 0-Star category.
- **Sampling Methodology:** I sampled and annotated 10 representative runs that marked major record improvements and were available on YouTube, ensuring a balanced historical coverage.

Annotation Results

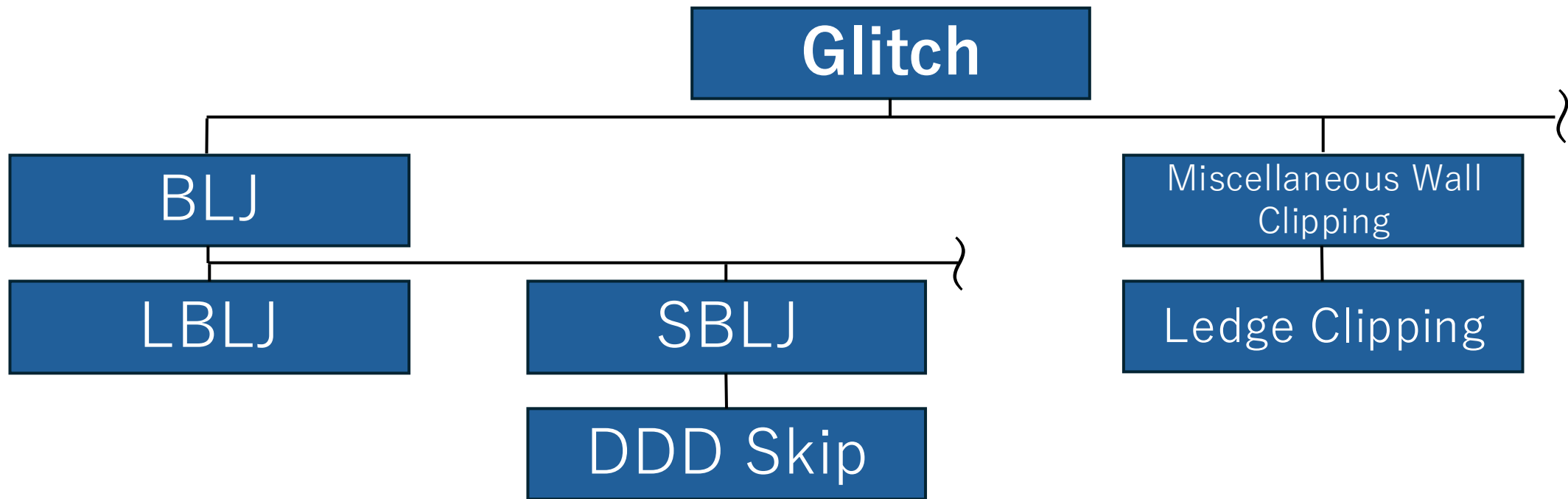
- Consistency in Glitch Types: As shown in the data, the types of glitches utilized have remained largely consistent over time (with minor shifts such as the adoption of the Pole Glitch and recent BLJ usage).
- Duration Patterns: Two primary patterns were observed regarding glitch durations: those that remain stable and those that are steadily decreasing.
- Interpretations: Reductions in frame counts are attributed to improved player proficiency, while occasional increases reflect the extreme execution difficulty that remains unstable even for elite players.

Is the current glitch list appropriate?

Some glitches are very similar and potentially confusing:

- BLJ: High-speed movement utilizing repeated long jumps.
- LBLJ: A specific variant of BLJ performed in the castle lobby.
- Miscellaneous Wall Clipping (MWC): Wall clipping achieved without using SM64-specific mechanics like BLJ.
- Ledge Clipping: A sub-type of MWC that exploits momentary gaps in collision hitboxes during ledge-grab or pull-up animations.

Our Proposal: Hierarchical Glitch List



- If the model identifies an LBLJ as a general BLJ, we would consider it a successful detection (OK) because the core event was accurately captured.

Alternative Approaches

- **Simplification:** Removing redundant labels to ensure a streamlined and minimal list.
- **Generalized Classification:** Adopting high-level categories like OoB (Out of Bounds) and SG (Speed Glitch), consistent with our previous research.
- **Taxonomy Alignment:** Following the taxonomy of prior work to maintain consistency with established research.

Key Question

- Given our goal of supporting the speedrun community, which of these approaches is the most appropriate?